

PERFORMANCE ANALYSIS OF UAV GPS SYSTEMS UNDER SPOOFING ATTACK USING DISCRETE MARKOV CHAINS

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ABSTRACT

Ensuring the reliability and security of GPS communication is critical for autonomous vehicle navigation. This article presents a Discrete-Time Markov Chain (DTMC)-based analysis of a public GPS spoofing dataset to evaluate the system's security posture. Using a transition probability matrix derived from the data's event sequence, the study models the behavior of a GPS receiver across safe and multiple spoofed states. Key metrics, including steady-state probabilities, system utilization, and mean recurrence time, are calculated to assess system resilience and vulnerability. The analysis reveals a system that is highly resilient, returning to a safe state in an average of 1.28 steps following an anomaly, yet remains vulnerable, with spoofing events present in over 22% of the observed data. Furthermore, the model uncovers two distinct behavioral profiles: fleeting, low-level attacks that resolve immediately and highly persistent sophisticated attacks that pose a more significant threat. Additionally, a sensitivity analysis was conducted to investigate the interaction between the stability and the system resilience, which demonstrated that the ability of recovery is more critical in low-stability scenarios. The DTMC framework provides rigorous, probabilistic insights into spoofing attack patterns, informing the design of more context-aware detection strategies to enhance GPS security under adversarial conditions.

Keywords: unmanned aerial vehicles, security, GPS, spoofing, Discrete-Time Markov Chain

1. INTRODUCTION

1.1 Background

Global Positioning System (GPS) technology is indispensable across various sectors, enabling accurate positioning and navigation for many vehicles and systems. Among these, drones—also known as unmanned aerial vehicles (UAVs)—are increasingly prevalent in military operations, commercial transport, delivery services, agriculture, infrastructure inspection, and recreational use by civilians. Unlike ground or maritime vehicles, drones rely heavily on GPS for precise flight control, navigation, and safety due to their three-dimensional mobility and lack of physical road constraints. This dependence makes drones particularly vulnerable to GPS spoofing attacks, where adversaries broadcast counterfeit signals to manipulate drone positioning or disrupt operations. The consequences can range from loss of control and collision risks to serious

security threats in defense and critical infrastructure contexts. Therefore, understanding and mitigating GPS spoofing threats in drones is critical to ensure safe and reliable UAV operations. *B. Motivation*

The increasing sophistication of GPS spoofing attacks demands advanced analytical tools to evaluate system performance and security. While real-world attack testing is often impractical due to legal and safety constraints, simulation-based datasets provide a safe alternative to study these threats [1]. However, analyzing the temporal dynamics of GPS signal states—safe or compromised—requires probabilistic models that can capture state transitions over time. Markov chains offer a powerful framework to model these transitions, enabling the quantification of system availability and the likelihood and timing of spoofing events.

1.2 Motivations

As drones become more ubiquitous and operate in increasingly complex environments, the sophistication and frequency of GPS spoofing attacks also rise [2]. Testing these attacks in real environments is constrained by legal and safety considerations, leading to reliance on simulation datasets for research. However, assessing the security and availability of GPS communication in drones requires detailed analysis of the signal state transitions—how the system moves between safe operation and attack states over time. Discrete-Time Markov Chains (DTMC) provide a robust probabilistic framework to model these temporal transitions, enabling insights into the long-run operational safety, the probability of transitioning into spoofed states, and the average number of steps required to recover from an attack. These insights are vital for designing more resilient drone navigation systems and developing effective countermeasures.

1.3 Objectives

This paper aims to:

1. Develop a DTMC model based on a GPS spoofing dataset to represent the state transitions between safe and spoofed GPS signals.
2. Calculate key performance metrics, including system utilization, average attack severity, and mean recurrence time for each state.
3. Analyze the calculated metrics to characterize the system's resilience, identify distinct attack profiles (e.g., fleeting vs. persistent), and provide a quantitative assessment of the system's vulnerabilities.
4. Conduct a sensitivity analysis to investigate how the long-term system's security is impacted by variations in its intrinsic stability and ability to recover from attacks.

The entire data processing and analysis was implemented in Python, and the source code is available at: <https://github.com/laiszig/uav-gps-spoofing-markov>

2. METHODOLOGY

2.1 Model Description

To analyze the sequential nature of spoofing events and the resilience of the GPS system, we model its behavior as a Discrete-Time Markov Chain (DTMC). A DTMC is a stochastic model that describes a sequence of events where the probability of moving to the next state depends only on the present state, not on the sequence of events that preceded it [3].

States

In the context of this model, a "state" represents the operational condition of the GPS receiver at each discrete time step. Let the random variable X_t denote the state of the system at step t . The state space S is defined by four distinct states, representing both a normal operational condition and three tiers of spoofing attack severity.

The state space is defined as $S = \{0, 1, 2, 3\}$, where each state corresponds to:

- State 0 (Safe): The system is operating normally, and the GPS signal is considered authentic.
- State 1 (Simplistic Attack): The system is affected by a low complexity spoofing attack.
- State 2 (Intermediate Attack): The system is affected by a medium complexity spoofing attack.
- State 3 (Sophisticated Attack): The system is affected by a high complexity, persistent spoofing attack.

Transition Probability Matrix (P)

The evolution of the system from one step to the next is governed by a set of transition probabilities. The transition probability p_{ij} is the probability that the system will move to state j at the next step ($t + 1$), given that it is currently in state i at step t . This is formally expressed as:

$$P_{ij} = P(X_{t+1} = j \mid X_t = i)$$

These probabilities are collected into a square matrix known as the Transition Probability Matrix (P). For our four-state system, the matrix is structured as follows:

$$P = \begin{bmatrix} p_{00} & p_{01} & p_{02} & p_{03} \\ p_{10} & p_{11} & p_{12} & p_{13} \\ p_{20} & p_{21} & p_{22} & p_{23} \\ p_{30} & p_{31} & p_{32} & p_{33} \end{bmatrix}$$

A fundamental property of this matrix is that the sum of the probabilities of transitioning out of any given state must be equal to 1. Therefore, for any row i in the matrix, $\sum_j p_{ij} = 1$.

Steady-State Probabilities (π)

After the system has run for a large number of steps, the probability of finding it in any particular state i becomes constant. This long-run probability distribution is known as the steady-state distribution. It is represented by a vector $\pi = [\pi_0, \pi_1, \pi_2, \pi_3]$.

Each element π_i represents the long-run proportion of steps that the system spends in state i . For example, π_0 is the probability of observing the system in a "Safe" state at any given step in the long run. The steady-state vector is a key indicator of the system's overall stability and vulnerability, and it is found by solving the characteristic equation $\pi P = \pi$, subject to the constraint that $\sum \pi_i = 1$.

2.2 System Configuration

This study utilizes a public dataset titled "A DATASET for GPS Spoofing Detection on Autonomous Vehicles" [1]. The dataset provides a rich collection of features extracted from Global Positioning System (GPS) signals, designed specifically for the development of machine learning-based detection techniques. The dataset contains 158,170 samples, which include both legitimate GPS signals and instances corresponding to three distinct types of simulated spoofing attacks.

The data was collected using a Universal Software Radio Peripheral (USRP) unit configured as an eight-channel GPS receiver, emulating the equipment on a real autonomous vehicle. The data collection process was conducted under two different scenarios to ensure diversity: a moving vehicle test with speeds ranging from 0 to 60 mph and a static test with the receiver placed in three stationary positions at varying altitudes. This setup allowed for the real-time extraction of 13 features across eight parallel channels at different stages of the signal processing chain.

Considering simplicity and relevance this study utilizes a simplified 2D format of the dataset instead of the original one containing raw output from all eight channels. The time series data from the parallel channels was converted into a two-dimensional feature map. Also, data points from time spans when

a channel was not locked onto a GPS signal—represented by zero values in the original dataset—were eliminated to streamline the data.

For this analysis, a specific subset of features was used to clean the data and define the system's state. The key features are:

Time of the Week (TOW): This feature represents the number of elapsed seconds since the beginning of the GPS week as determined by the satellite's atomic clock.

Receiver Time (RX): This is the local receiver's time in seconds, which starts from the beginning of the TOW.

Satellite Vehicle Number (PRN): Each GPS satellite is identified by a unique number, referred to as its PRN.

Carrier-to-Noise Ratio (C/N): Provided in the dataset as CN0, this is the ratio of the received carrier signal strength to the received noise strength and serves as a key indicator of signal quality.

Output: This is a label field within the dataset used to separate the signals into distinct classes. For this analysis, the Output column defines the system's state, indicating whether the signal is legitimate (State 0) or corresponds to one of the three simulated spoofing attack types (States 1, 2, and 3).

3. IMPLEMENTATION

3.1 Data Cleaning and Preprocessing

The source dataset, while comprehensive, requires a structured preprocessing workflow to be suitable for Markov chain analysis. The primary challenge lies in the presence of multiple data entries for single timestamps (TOW), a result of the eight-channel receiver simultaneously capturing signals from different satellites. The objective of this preprocessing phase was to distill this complex data into a single, unambiguous, and chronologically ordered sequence of system states.

Only the columns essential for the analysis were loaded: TOW, RX, PRN, CN0, and Output. A custom logic to resolve the duplicate timestamp entries was developed, and to ensure that each timestamp in the final dataset represented the most reliable information available, a hierarchical selection process was implemented. The primary criterion for selection was signal quality, as

measured by the Carrier-to-Noise Ratio (CN0). For each group of entries sharing an identical TOW value, the row with the highest CN0 was selected as the most representative measurement for that moment.

A tie-breaking rule was established for the specific case where multiple entries for the same timestamp shared the same highest CN0 value. To ensure that events of security interest were prioritized, the entry with the highest Output state value was retained. This logic guarantees that if a "Safe" (State 0) and an "Attack" (State 1, 2, or 3) signal are recorded with the same signal strength, the attack state is preserved in the final dataset, which is crucial for a security-focused performance analysis. This entire selection process was implemented efficiently through a multi-level sort of the data, followed by a command to remove duplicates while retaining only the first, most relevant entry for each timestamp. The result of this procedure is a clean dataset where each row corresponds to a unique and validated system state, providing a reliable foundation for the subsequent DTMC analysis.

3.2 Parameter Estimation

While the dataset contains high-resolution timestamps, a Discrete-Time analysis was selected as the most appropriate model for this paper. The rationale for treating time as a series of discrete steps rather than a continuous variable stems from the nature of the system being studied. The primary interest of this research is not the specific *duration* of spoofing attacks, but rather the *sequential dynamics* of the system's state transitions. Given that the objective of many spoofing attacks is the rapid acquisition of sensor data, the events are often fleeting. A step-based analysis, where each row of the cleaned dataset represents a distinct state observation, better captures the reactive, step-by-step behavior of the system as it transitions between safe and various attack states.

The central parameter estimated for the DTMC model is the Transition Probability Matrix (P). This matrix quantitatively describes the system's evolution from one step to the next. The estimation process was based directly on the empirical data from the cleaned and chronologically sorted dataset.

First, a transition count matrix was constructed by traversing the dataset sequentially. For each step t

(from the second step to the end), the starting state, X_{t-1} , and the ending state, X_t , were identified. A counter for that specific $i \rightarrow j$ transition was then incremented. This was repeated for the entire dataset to obtain the total number of observed transitions between every pair of states.

Next, these raw counts were converted into probabilities to populate the P-matrix. The probability p_{ij} of transitioning from a starting state i to an ending state j was calculated by normalizing the transition count by the total number of times the system was observed in the starting state i . The formula used is:

$$p_{ij} = \frac{N_{ij}}{N_i}$$

Where N_{ij} is the total observed count of transitions from state i to state j , and N_i is the total number of steps the system spent in state i . This method provides a maximum likelihood estimate of the transition probabilities, resulting in a P-matrix that accurately reflects the behavior observed in the dataset.

3.3 Performance Metric Calculation

With the Transition Probability Matrix (P) estimated from the data, a set of performance metrics was derived to quantitatively evaluate the system's behavior. These metrics are calculated based on the steady-state distribution of the Markov chain, which describes the long-run behavior of the system.

Steady-State Probabilities (π)

The steady-state probability vector, $\pi = [\pi_0, \pi_1, \pi_2, \pi_3]$, represents the long-run proportion of steps the system is expected to spend in each state. This vector was calculated by solving the characteristic eigenvector equation for a Discrete-Time Markov Chain:

$$\pi P = \pi$$

This equation signifies that in the steady state, the probability distribution across the states remains unchanged after one transition step. To find a unique solution for π , this system of linear equations is solved under the additional constraint that the sum of all steady-state probabilities must equal one:

$$\sum \pi_i = 1$$

Solving this system yields the long-run probability for each system state, which forms the foundation for calculating the subsequent performance metrics.

System Utilization (U)

Utilization is defined as the proportion of steps that the system spends in any of the attack states (States 1, 2, or 3). It serves as a direct measure of the system's overall vulnerability and exposure to spoofing threats. It is calculated by summing the steady-state probabilities of all non-safe states, or more simply by subtracting the probability of being in the safe state from one:

$$U = 1 - \pi_0$$

Average Attack Severity (N)

To represent the system's overall risk level with a single value, the Average Attack Severity was calculated. This metric is a weighted average of the system's states, where each state is weighted by its severity level (i) and its long-run probability (π_i). A higher value indicates that the system tends to spend more time in more severe attack states. The formula is:

$$N = \sum(i * \pi_i) = (0 * \pi_0) + (1 * \pi_1) + (2 * \pi_2) + (3 * \pi_3)$$

Mean Recurrence Time (W)

The Mean Recurrence Time, W_i , measures the average number of steps required for the system to return to a state i after it has left that state. This metric is particularly useful for quantifying the system's resilience and recovery speed. For example, W_0 indicates how quickly the system returns to a "Safe" state after an attack has occurred. For any state i , the Mean Recurrence Time is the reciprocal of its steady-state probability:

$$W_i = 1 / \pi_i$$

4. RESULTS

The application of the data processing and analysis methodology described in the previous section yielded the quantitative results presented below. These findings characterize the behavior of the GPS system based on the analyzed dataset.

4.1 Transition Probabilities

The parameter estimation process resulted in the Transition Probability Matrix (P) represented in Table 1. Each element p_{ij} represents the probability of the system transitioning from a starting state i to an ending state j in the subsequent time step.

The matrix provides a quantitative summary of the system's step-by-step dynamics as observed in the dataset. Notably, the diagonal elements indicate the probability of the system remaining in the same state. A high probability of self-transition was observed for the Safe state ($P_{00} = 0.8310$) and the Sophisticated attack state ($p_{33} = 0.6071$). For the lower-level attack states, the most probable transition was a return to the Safe state, with probabilities of $p_{10} = 0.7571$ and $p_{20} = 0.6313$. The matrix also reveals that certain transitions, such as from State 1 or State 2 to State 3, were not observed in the dataset and thus have a probability of zero.

4.2 System Performance Metrics

Based on the Transition Probability Matrix, the long-run steady-state distribution and key performance metrics for the system were calculated.

1) *Steady-State Probabilities (π)*

The steady-state probabilities (π), representing the long-run proportion of steps the system is expected to spend in each state, were calculated as follows:

$\pi(0)$ - Safe: 0.7785 (77.85%)

$\pi(1)$ - Simplistic: 0.0633 (6.33%)

$\pi(2)$ - Intermediate: 0.0902 (9.02%)

$\pi(3)$ - Sophisticated: 0.0680 (6.80%)

System Utilization (U)

The total probability of the system being in any of the attack states (1, 2, or 3) is defined as the System Utilization (U). $U = 0.2215$ indicates the system is in an attack state for 22.15% of the observed steps.

Average Attack Severity (N)

While the System Utilization metric of 22.15% reveals that spoofing events are frequent, the Average Attack Severity (N) of 0.4477 provides a more nuanced view of the system's overall risk posture. This low value, on a scale from 0 (Safe) to 3 (Sophisticated), indicates that the system's

operational 'center of gravity' remains heavily anchored to the Safe state. It demonstrates that the frequent attacks observed in the dataset are predominantly of lower severity, preventing the overall risk profile from escalating to more dangerous levels.

Mean Recurrence Time (M)

The Mean Recurrence Time (W) represents the average number of steps it takes for the system to return to a specific state after leaving it.

State	Average Steps to Return
State 0	1.28 steps
State 1	15.79 steps
State 2	11.09 steps
State 3	14.70steps

State Transition Diagram

To visually represent the dynamics of the system, the Transition Probability Matrix was used to generate the state transition diagram shown in Figure 1. The diagram illustrates the states as nodes and the transition probabilities as labeled, directed edges between them.

4.3 Sensitivity Analysis

To further understand the system's dynamics, a sensitivity analysis was conducted. This complementary study moves beyond the single scenario observed in the dataset and explores, through iterative simulations, how the system's long-term safety (the steady-state probability of being in the Safe state, π_0) responds to variations in two critical parameters: its intrinsic stability (the probability p_{00} of remaining in the Safe state) and its recovery resilience (the probability of returning to the Safe state after an attack). The objective is to identify which of these factors has a greater impact on the system's overall security under different conditions. The results are visualized in Figure 3, which plots π_0 as a function of p_{00} for ten different resilience scenarios.

The correlation between p_{00} and π_0 exhibits a sigmoidal characteristic, where the impact of increasing p_{00} is most pronounced in the mid-range values. The level of recovery resilience is shown to

be a significant factor. For example, in the highest resilience scenario (100% of observed recovery capability), π_0 ranges from 39.76% to 98.34%. In contrast, for the lowest resilience scenario (10% of recovery capability), the values for π_0 are substantially lower, ranging from 6.24% to 85.68% over the same variation of p_{00} . It is also observed that the performance gap between the different resilience scenarios is most pronounced at lower levels of intrinsic stability and diminishes as π_{00} approaches 1.0.

Table 1 Transition Probability Matrix (P)

	To State 0	To State 1	To State 2	To State 3
From State 0 (Safe)	0.8310	0.0568	0.0779	0.0343
From State 1 (Simplistic)	0.7571	0.0775	0.1655	0.0000
From State 2 (Intermediate)	0.6313	0.1574	0.2113	0.0000
From State 3 (Sophisticated)	0.3929	0.0000	0.0000	0.6071

Long-Run Probability of System States

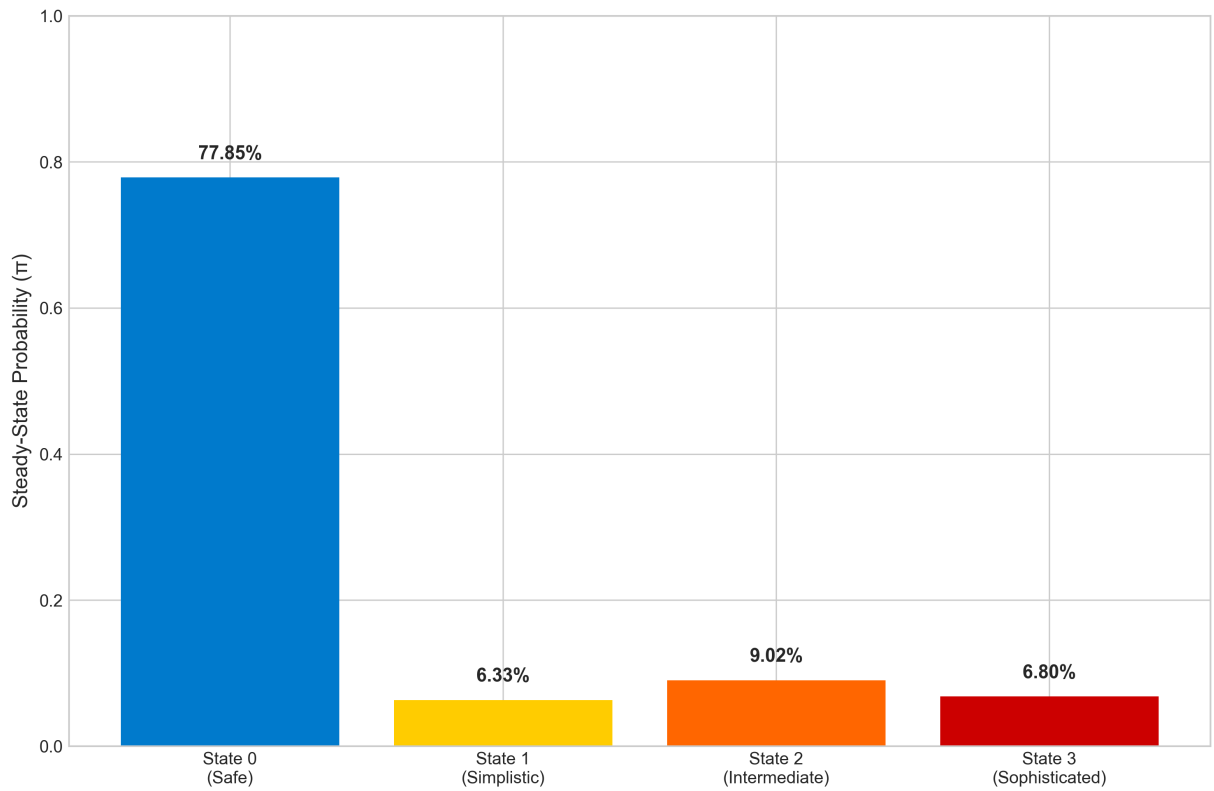


Figure 1 Steady-State Probabilities (π).

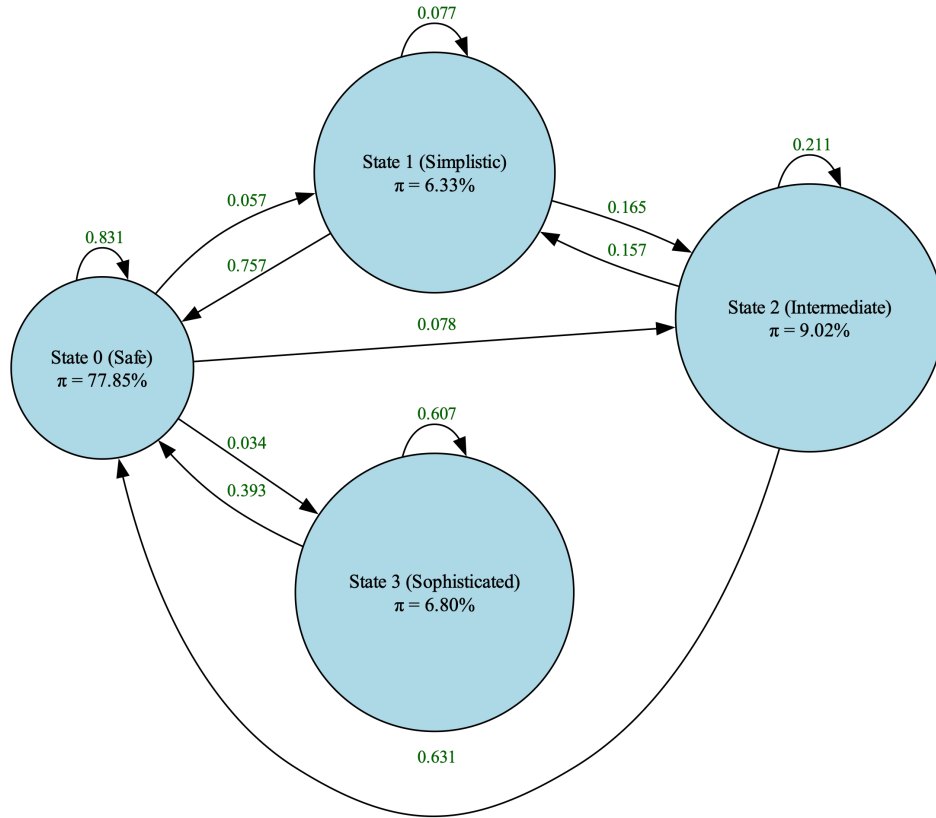


Figure 2 DTMC State Transition Diagram for GPS System States. The diagram shows the probability of transitioning from one state to another at each step. The steady-state probability (π) for each state is also noted within the node.

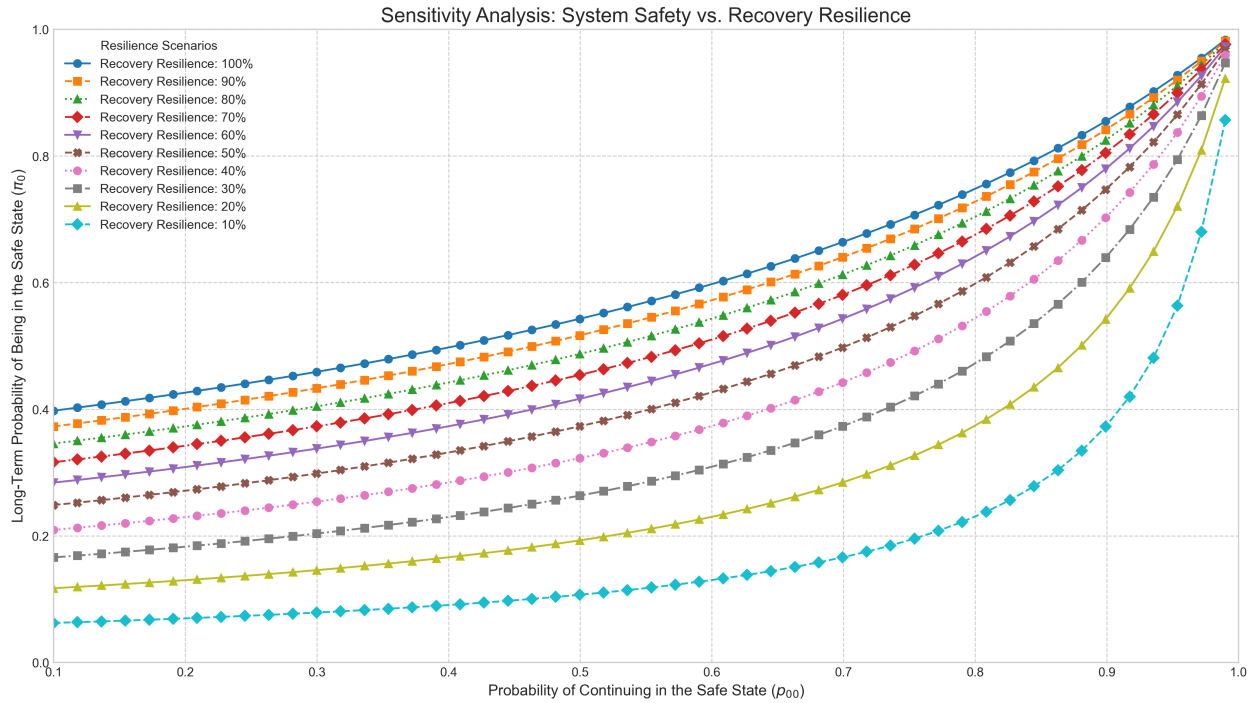


Figure 3 Sensitivity Analysis of Long-Term System Safety (π_0) as a function of Intrinsic Stability (p_{00}) under various Recovery Resilience scenarios.

5. DISCUSSION

The results of the Discrete-Time Markov Chain (DTMC) analysis provide a quantitative characterization of the event sequences within the GPS spoofing dataset. Before interpreting the metrics, it is crucial to situate them within the context of the dataset's original purpose. The dataset was not created as a continuous operational log but as a curated collection of events designed to train machine learning models to identify attacks. Therefore, the "time" between each step in our DTMC model is not uniform real-world time, but a transition between discrete, recorded events. This context is key to understanding the findings.

5.1 Patterns of Recovery and Event Sequencing

The analysis reveals a strong pattern of rapid return to a "Safe" state following an attack event. The Mean Recurrence Time for State 0 (W_0) of just 1.28 steps indicates that after a spoofing event is recorded, the very next event is highly likely to be a safe one. This observed "recovery" could be interpreted in two ways:

1. It may reflect the GPS receiver's inherent resilience and its ability to quickly re-acquire authentic signals after a brief disruption.
2. Alternatively, and perhaps more likely given the dataset's structure, it may reflect the nature of the data collection itself, where discrete attack scenarios were recorded as short-lived events. The quick return to "Safe" could be the result of the spoofing signal being intentionally stopped by the experimenter or a simulated system reboot after an anomaly was logged.

Regardless of the cause, the practical implication is that isolated, non-persistent attack events are the norm in the dataset.

5.2 Quantifying Attack Event Frequency

The System Utilization (U) of 22.15% is a significant finding. Within the context of this dataset, it means that over one-fifth of all recorded events are classified as an attack state. This highlights that the dataset is rich with the very instances needed to train a robust detection model, as spoofing events are well-represented. The steady-state probabilities (π) further reveal that

"Intermediate" attacks (State 2) are the most frequently recorded type of anomaly (9.02% of all events).

5.3 Characterizing Attack Profiles by Sequential Persistence

The most critical insight provided by the DTMC analysis is the clear distinction between different attack profiles based on their sequential behavior, as seen in the Transition Probability Matrix (P).

Fleeting Attack Events: Simplistic (State 1) and Intermediate (State 2) attacks demonstrate a clear lack of persistence. If the current event is a State 1 or 2 attack, there is a very high probability (75.7% and 63.1%, respectively) that the next recorded event will be a return to the Safe state. These appear as isolated, "one-off" events in the data sequence.

Persistent Attack Sequences: In stark contrast, the Sophisticated (State 3) attack exhibits high persistence. The probability of a State 3 event being followed by another State 3 event is 60.7% (p_{33}). This suggests that these attacks are not recorded as isolated incidents but as a persistent sequence of events.

This finding is paramount for designing detection systems. An ML model should not only classify a single event but also consider the sequence. A transition to State 1 or 2 is a low-risk anomaly that is likely to resolve, but a transition to State 3 indicates the start of a potentially sustained and far more dangerous event sequence.

5.4 Sensitivity Analysis – The Impact of Stability and Resilience

An analysis of the results, presented in Figure 3, reveals three key conclusions. First, as expected, for any given level of resilience, increasing the system's intrinsic stability (p_{00}) results in a corresponding increase in long-term safety (π_0). This relationship follows a distinct sigmoid or "S-curve," indicating diminishing returns at the extremes; the most significant gains in long-term safety occur when improving a system with moderate stability.

Second, a horizontal comparison across the curves reveals the fundamental role of recovery resilience. For a system with moderate stability (e.g., $p_{00} \approx 0.52$), there is a vast difference in outcomes. A highly resilient system (100% of observed recovery

capability) remains in the Safe state 55.39% of the time, whereas a system with low resilience (10% of recovery capability) is safe only 11.12% of the time. This demonstrates that a rapid recovery mechanism is an extremely effective defense, particularly for systems that are not inherently perfect.

The most significant insight, however, is the synergistic relationship between these two factors. The analysis shows that recovery resilience is most critical when the system is inherently unstable. At low values of p_{00} (e.g., 0.24), the gap between the curves is enormous; a highly resilient system can maintain a long-term safety level of 43.89%, while a system with low resilience collapses to just 7.30%. This suggests that a strong recovery capability can effectively safeguard an otherwise unstable system. Conversely, high intrinsic stability can compensate for poor recovery resilience. At high values of p_{00} (e.g., 0.99), the curves converge, and the performance gap narrows significantly. A highly stable system maintains a safety level of 98.34% with high resilience, and still achieves 85.68% even with the lowest resilience tested. If a system rarely enters an attack state, its ability to recover from those infrequent events becomes less critical to its overall long-term performance.

In summary, this sensitivity analysis demonstrates that the long-term security of the GPS system is not dependent on a single factor but on the interplay between its ability to remain safe and its ability to recover. Systems prone to disruption are critically dependent on rapid recovery, whereas highly stable systems are robust even with slower recovery mechanisms.

6. CONCLUSION

This paper set out to evaluate the performance and behavioral dynamics of a GPS system under various spoofing attacks by applying a stochastic modeling approach. Leveraging a public dataset specifically designed for spoofing detection research, we developed and applied a Discrete-Time Markov Chain (DTMC) model to analyze the sequential patterns of transitions between a "Safe" operational state and three distinct tiers of attack severity. This method proved appropriate for capturing the stochastic, step-by-step nature of the events recorded in the dataset.

The analysis revealed that the sequence of events within the dataset is characterized by a rapid return to the "Safe" state following most anomalies. However, a more nuanced look at the transition probabilities uncovered two distinct attack profiles: lower-level "fleeting" attacks that are typically isolated events, and "persistent" sophisticated attacks that are highly likely to appear in a sustained sequence. The key conclusion is not necessarily a measure of the receiver's real-time resilience, but rather a characterization of the threat patterns present in the data. The system is exposed to attack events in over 22% of the observed steps, and the nature of that threat varies significantly by type. Furthermore, a sensitivity analysis demonstrated the synergistic relationship between the system's intrinsic stability and its recovery capability, revealing that a rapid recovery mechanism is most critical for safeguarding an inherently unstable system.

These findings have direct implications for the design of intelligent detection systems. By understanding these sequential patterns, a security model can be developed that moves beyond simple state classification to a more context-aware risk assessment. For example, the detection of a "Sophisticated" attack event can be assigned a much higher alert level, as it is likely to persist.

Future work could extend this analysis in several directions. Applying this DTMC modeling methodology to other GPS datasets could determine if these "fleeting" versus "persistent" attack patterns are a universal characteristic. Furthermore, the transition probabilities calculated in this study could form the basis of a predictive, real-time detection engine that dynamically assesses the threat level based not just on the system's current state, but on the probability of it transitioning to or remaining in a more dangerous state. Finally, further sensitivity analyses could be performed to explore the impact of other parameters, such as the probabilities of attack escalation.

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